

The RG-invariant entropy-maximal variety Σ : fixed points, orbit collapse, and the hypertetrahedral component at $N = 4$

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Abstract

We adopt the natural definition of the RG-invariant entropy-maximal variety Σ as the *maximal subset of the entropy-max locus* $\mathcal{M}_{1|N} = \{T = 0\}$ that is preserved by the RG flow, i.e. the common solution of $T^{(k)} = 0$ for all $k \geq 0$, allowing arbitrarily many connected components. Under this definition every RG fixed point lying on $\mathcal{M}_{1|N}$ is automatically a component of Σ , because at a fixed point $\beta = 0$ forces all flow-derivatives $T^{(k \geq 1)}$ to vanish identically. The earlier “maximal linear RG-invariant subspace” Σ_N^{lin} is then just one connected component of Σ — the largest linear, group-symmetric one — and Σ in general contains additional isolated fixed-point orbits. With this framing we re-examine $N = 3, 4$. At $N = 3$, modulo the gauge redundancy $O(3)$, Σ has no fixed-point component beyond Σ_T^{lin} : the apparent “rogue” fixed points P_2, P_3 are the cubic and decoupled fixed points in a rotated basis, and the claim $S_{2|2}(P_2) > S_{2|2}(\text{Heis})$ is a basis artefact ($S_{2|2}$ is not $O(N)$ invariant). At $N = 4$, Σ acquires a genuinely new component: the hypertetrahedral $S_5 \times \mathbb{Z}_2$ fixed point, a distinct $O(4)$ orbit (proved via the $O(4)$ -invariant spectrum of the Sym^2 flattening) that does not lie in Σ_4^{lin} even after any rotation. We give its explicit 4-simplex construction.

1 Definition of Σ and the fixed-point inclusion

For the fully symmetric quartic coupling $\lambda_{ijkl} = \lambda_{(ijkl)}$ on $\mathbb{R}^N$ define the $1|N$ reduced density matrix and its traceless part

$$(\rho'_{1|N})_{ij} = \sum_{klm} \lambda_{iklm} \lambda_{jklm}, \quad T_{ij} = (\rho'_{1|N})_{ij} - \frac{1}{N} \text{Tr}(\rho'_{1|N}) \delta_{ij}, \quad (1)$$

so the entropy-maximal locus is $\mathcal{M}_{1|N} = \{T = 0\}$, and the one-loop beta function (units $\kappa = \epsilon = 1$) is

$$\beta_{ijkl} = -\lambda_{ijkl} + \sum_{s,t,u \text{ channels}} \lambda_{ijab} \lambda_{abkl}. \quad (2)$$

The flow derivatives of T are $T^{(0)} = T$ and $T^{(k)} = \left(\sum_a \beta_a \partial_{\lambda_a}\right)^k T$.

Definition 1 (RG-invariant entropy-maximal variety).

$$\Sigma := \{\lambda : T^{(k)}(\lambda) = 0 \text{ for all } k \geq 0\}.$$

Equivalently, Σ is the largest RG-flow-invariant subset of $\mathcal{M}_{1|N}$: the flow line through any point of Σ stays in $\mathcal{M}_{1|N}$ for all RG time. Connectedness is not assumed — Σ may be a union of components of different dimensions.

Proposition 1 (Fixed points are in Σ). *Every RG fixed point λ_* lying on $\mathcal{M}_{1|N}$ satisfies $\lambda_* \in \Sigma$.*

Proof. By definition $T^{(0)}(\lambda_*) = T(\lambda_*) = 0$ since $\lambda_* \in \mathcal{M}_{1|N}$. For $k \geq 1$, each $T^{(k)} = (\beta \cdot \partial)^k T$ is a sum of monomials *every one of which carries at least one explicit factor of β* (differentiating along the flow always introduces $d\lambda/dt = \beta$). At a fixed point $\beta(\lambda_*) = 0$, so every such monomial vanishes and $T^{(k)}(\lambda_*) = 0$ for all $k \geq 1$. Hence λ_* satisfies all the defining equations of Σ . $\square$

Remark 1. *Requiring all $T^{(k)} = 0$ (not merely $T = 0, T^{(1)} = 0$) is what makes Σ genuinely flow-invariant rather than first-order tangent. In practice the chain $T^{(0)} = T^{(1)} = \dots$ generates an ideal that becomes flow-stable after finitely many steps, so Σ is computable. The old “maximal linear subspace” $\Sigma_N^{\text{lin}} = \{\lambda_{iiii} = u, \lambda_{iijj} = w, \text{else} = 0\}$ is the largest linear, finite-group-symmetric component of Σ ; we verified for $N = 2, \dots, 6$ that $T^{(0)} = T^{(1)} = T^{(2)} \equiv 0$ on Σ_N^{lin} with the reduced flow $\dot{u} = -u + 3u^2 + 3(N-1)w^2$, $\dot{w} = -w + 2uw + (N+2)w^2$.*

The structural question becomes: *what are the components of Σ , modulo the gauge redundancy $O(N)$?* A fixed point of β on $\mathcal{M}_{1|N}$ is a zero-dimensional component (an isolated orbit); Σ_N^{lin} is a two-dimensional component. We now enumerate the fixed-point components at $N = 3, 4$.

2 $N = 3$: Σ has no fixed-point component beyond Σ_T^{lin}

2.1 The S_3 sector

In the permutation-symmetric ansatz with orbit amplitudes $A = \lambda_{iiii}$, $B = \lambda_{iiij}$, $C = \lambda_{iijj}$, $D = \lambda_{iijk}$, the flow is

$$\begin{aligned}\dot{A} &= -A + 3A^2 + 12B^2 + 6C^2 + 6D^2, \\ \dot{B} &= -B + 3AB + 9BC + 6BD + 3CD + 6D^2, \\ \dot{C} &= -C + 2AC + 6B^2 + 4BD + 5C^2 + 10D^2, \\ \dot{D} &= -D + AD + 2B^2 + 2BC + 8BD + 10CD + 4D^2.\end{aligned}\tag{3}$$

Solving $\beta = 0$ and intersecting with $\mathcal{M}_{1|3}$ (in this sector a single equation $F = 2AB + B^2 + 6BC + 6BD + 6CD + 6D^2 = 0$) gives six real fixed points on $\mathcal{M}_{1|3}$:

name	(A, B, C, D)	in Σ_T^{lin} ?	$S_{2 2}$	$O(3)$ orbit
Gaussian	$(0, 0, 0, 0)$	yes	—	trivial
Decoupled	$(1/3, 0, 0, 0)$	yes	1.0986	Decoupled
Cubic	$(2/9, 0, 1/9, 0)$	yes	1.1564	Cubic
Heisenberg	$(3/11, 0, 1/11, 0)$	yes	1.2092	Heisenberg
P_2	$(70, -2, 19, 4)/243$	no	1.2113	Cubic
P_3	$(11, 2, 8, -4)/81$	no	1.0566	Decoupled

By Proposition 1 all six are components of Σ . The question is how many *distinct* $O(3)$ orbits they represent.

2.2 The “rogue” points are rotated copies

The full $O(3)$ -invariant fingerprints ($\|\lambda\|^2, a_0, a_1, \text{Tr } \rho^2, \text{Tr } \rho^3$) of P_2 and the cubic fixed point coincide identically $(\frac{10}{27}, \frac{4}{3}, \frac{16}{27}, \frac{100}{2187}, \frac{1000}{177147})$, and P_3 matches the decoupled fixed point. We found explicit $R \in O(3)$ with $\lambda_{\text{Cubic}} \xrightarrow{R} \lambda_{P_2}$ and $\lambda_{\text{Decoupled}} \xrightarrow{R} \lambda_{P_3}$ to residual distance $\sim 10^{-14}$.

Proposition 2. *At $N = 3$, modulo $O(3)$ the fixed-point components of Σ are exactly the four classical orbits Gaussian, Decoupled, Cubic, Heisenberg, all of which already have a representative inside Σ_T^{lin} . The “rogue” points P_2, P_3 are not new components — they are the cubic and decoupled orbits in a rotated flavour basis.*

2.3 The 2|2 entropy inequality is a basis artefact

The quantity $S_{2|2}$ is computed from $\rho'_{2|2}$ in a *fixed* flavour basis and is therefore not an $O(3)$ invariant: the same orbit gives different $S_{2|2}$ in different orientations. The cubic orbit reads $S_{2|2} = 1.1564$ when axis-aligned but 1.2113 in the P_2 orientation; the increase is pure orientation, with no invariant meaning. The earlier inference “ P_2 beats Heisenberg” ($1.2113 > 1.2092$) should therefore be withdrawn. Once orbits are compared invariantly (e.g. symmetrising ρ' over the orbit, or via $O(3)$ invariants) the Heisenberg orbit remains the genuine maximiser among the non-trivial orbits.

3 $N = 4$: Σ acquires the hypertetrahedral component

The S_4 sector (five amplitudes a, b, c, d, e) has five real fixed points on $\mathcal{M}_{1|3}$: Gaussian, the $SO(4)$ singlet $(a, c) = (\frac{1}{4}, \frac{1}{12})$, the tetrahedral-chiral point $(\frac{1}{12}, \frac{1}{12}, \frac{1}{12})$, and the Galois pair Twisted A/B. Their $O(4)$ fingerprints:

orbit	$\ \lambda\ ^2$	$a_0 = \lambda_{iijj}$
$SO(4)$ singlet	1/2	2
Hypertetrahedral (Twisted)	220/441	40/21
Decoupled = Tetra-chiral	4/9	4/3

Twisted A and B are the same orbit (rotation distance 1.6×10^{-13}); the tetrahedral-chiral point is the decoupled orbit (1.8×10^{-13}). The crucial difference from $N = 3$ is that the hypertetrahedral orbit is *not* reducible to the $SO(4)$ singlet or to Σ_4^{lin} (Section 4). Hence:

Proposition 3. *At $N = 4$, modulo $O(4)$, the fixed-point components of Σ are Gaussian, Decoupled, $SO(4)$ singlet, and the hypertetrahedral $S_5 \times \mathbb{Z}_2$ orbit. The first three meet Σ_4^{lin} ; the hypertetrahedral component lies entirely outside it. Thus Σ at $N = 4$ is strictly larger than the linear component Σ_4^{lin} :*

$$\Sigma = \Sigma_4^{\text{lin}} \cup \{\text{Gaussian}\} \cup \{\text{hypertetrahedral orbit}\} \cup \dots$$

where $\dots$ denotes any further non-fixed-point flow-invariant pieces.

4 What the hypertetrahedral component *is*, explicitly

4.1 Definition via the 4-simplex

Let $\{e^a\}_{a=1}^5 \subset \mathbb{R}^4$ be the five vertices of a regular 4-simplex (5-cell), centred and normalised so that

$$\sum_{a=1}^5 e^a = 0, \quad e^a \cdot e^b = \delta^{ab} - \frac{1}{5}. \quad (4)$$

The symmetry group of this configuration is S_5 acting on $\mathbb{R}^4$ through its standard 4-dimensional irrep; with the overall sign flip $\phi \mapsto -\phi$ this is $S_5 \times \mathbb{Z}_2 \subset O(4)$. The Molien series gives *two* invariant

quartics, so the general $S_5 \times \mathbb{Z}_2$ -invariant quartic is

$$V_{\text{hT}}(\phi) = u(\phi^2)^2 + v\tau(\phi), \quad \tau(\phi) := \sum_{a=1}^5 (e^a \cdot \phi)^4. \quad (5)$$

The first term is the $SO(4)$ singlet; the second, with tensor form $\lambda_{ijkl}^{(\tau)} = \sum_a e_i^a e_j^a e_k^a e_l^a$, is the new hypertetrahedral invariant. Unlike anything in Σ_4^{lin} it carries a nonzero fully-traceless rank-4 (25 of $SO(4)$) component.

4.2 The fixed point in the canonical S_4 basis

Rotated into the canonical basis where $S_4 \ltimes \mathbb{Z}_2^4$ is manifest, the hypertetrahedral fixed point is one of the two “Twisted” Galois conjugates, with S_4 -orbit amplitudes $(a, b, c, d, e) = (\lambda_{iiii}, \lambda_{iiij}, \lambda_{iijj}, \lambda_{iijk}, \lambda_{ijkl})$ given exactly by

$$(a, b, c, d, e) = \left(\frac{41 - 3\sqrt{5}}{168}, -\frac{1}{84}, \frac{13 + \sqrt{5}}{168}, \frac{1}{84}, \frac{-1 - \sqrt{5}}{168} \right). \quad (6)$$

One checks *exactly* that this point satisfies both defining conditions of Σ :

$$\rho'_{1|3} = \frac{55}{441} \mathbb{K}_4 \Rightarrow T = 0, \quad \beta_a = \beta_b = \beta_c = \beta_d = \beta_e = 0, \quad (7)$$

and consequently (Prop. 1) $T^{(k)} = 0$ for all k — so it is a bona fide component of Σ . Note $b, d, e \neq 0$, whereas every point of

$$\Sigma_4^{\text{lin}} = \{\lambda_{iiii} = u, \lambda_{iijj} = w, \text{ all other components} = 0\} \quad (8)$$

has $b = d = e = 0$. So in this basis it lies outside Σ_4^{lin} ; the substantive question is whether some $O(4)$ rotation brings it in.

4.3 Proof that it is a distinct component (outside Σ_4^{lin} mod $O(4)$)

Coarse invariants do not settle this. On Σ_4^{lin} , $\|\lambda\|^2 = 4u^2 + 36w^2$, $a_0 = 4u + 12w$, $a_1 = 4u^2 + 24uw + 36w^2$, and real (u, w) (involving $\sqrt{10}$) reproduce the hypertetrahedral values $(\frac{220}{441}, \frac{40}{21}, \frac{400}{441})$; the rank-4 norm also coincides. A *complete* discriminant is the spectrum of the $\text{Sym}^2(\mathbb{R}^4)$ flattening $Q_{(ij),(kl)} := \lambda_{ijkl} \in \mathbb{R}^{10 \times 10}$. Since $O(4)$ acts on $\text{Sym}^2(\mathbb{R}^4)$ by the induced representation, the *eigenvalue spectrum of Q is an $O(4)$ invariant*:

component	spec(Q) (with multiplicities)	pattern
$SO(4)$ singlet	$\frac{1}{6}^{(9)}, \frac{1}{2}^{(1)}$	$9 + 1 = \mathbf{9} \oplus \mathbf{1}$
Σ_4^{lin} , branch 1	$0.0583^{(3)}, 0.2089^{(6)}, 0.4762^{(1)}$	$3 + 6 + 1$
Σ_4^{lin} , branch 2	$0.1085^{(6)}, 0.2591^{(3)}, 0.4762^{(1)}$	$6 + 3 + 1$
hypertetrahedral	$0.0952^{(5)}, 0.2381^{(4)}, 0.4762^{(1)}$	$5 + 4 + 1$

The hypertetrahedral spectrum differs from every point of Σ_4^{lin} in both eigenvalues and degeneracy pattern. As an $O(4)$ invariant it cannot be changed by rotation, so:

Proposition 4. *The hypertetrahedral $S_5 \times \mathbb{Z}_2$ fixed point (6) is a component of Σ that is not $O(4)$ -conjugate to any point of Σ_4^{lin} . It is a genuinely distinct orbit on $\mathcal{M}_{1|3}$.*

The degeneracy pattern is the symmetry signature: $5 + 4 + 1$ is $\text{Sym}^2(\mathbf{4}) = \mathbf{1} \oplus \mathbf{4} \oplus \mathbf{5}$ for the standard S_5 irrep; the Σ_4^{lin} points carry the hypercubic $\mathbf{1} \oplus (3/6 \text{ split})$ of $S_4 \ltimes \mathbb{Z}_2^4$; the $SO(4)$ singlet carries $\mathbf{1} \oplus \mathbf{9}$. The three actions on $\text{Sym}^2(\mathbb{R}^4)$ are inequivalent, which is why no rotation relates them.

5 Status

	$N = 3$	$N = 4$
fixed-point orbits in $\Sigma \pmod{O(N)}$	Gaussian, Decoupled, Cubic, Heisenberg	Gaussian, Decoupled, $SO(4)$ singlet, Hypertetrahedral
all meet the linear component Σ_N^{lin} ?	yes	no
Σ strictly larger than Σ_N^{lin} ?	no	yes

Take-aways.

1. With Σ defined as the maximal flow-invariant subset of $\mathcal{M}_{1|N}$ (all $T^{(k)} = 0$, components and all), every RG fixed point on $\mathcal{M}_{1|N}$ is a component of Σ (Prop. 1). The old linear subspace Σ_N^{lin} is the largest linear, group-symmetric component.
2. At $N = 3$, modulo $O(3)$, Σ has no fixed-point component beyond Σ_T^{lin} . The “rogue” P_2, P_3 are the cubic/decoupled orbits in rotated bases, and the $S_{2|2}$ inequality is a non-invariant artefact.
3. At $N = 4$, Σ acquires a genuinely new component: the hypertetrahedral $S_5 \times \mathbb{Z}_2$ orbit, distinct from Σ_4^{lin} by the rotation-proof Sym^2 spectrum (Prop. 4).
4. Any entropy used to compare components must first be made $O(N)$ invariant (e.g. symmetrising ρ' over the orbit); orientation alone otherwise changes the verdict.

Method note. Fixed points were obtained by exact solution in each symmetry sector, cross-checked against multi-start Newton sweeps in the full coupling space; $T = 0$ was imposed exactly; the fixed-point inclusion in Σ is Proposition 1; orbit identifications used $O(N)$ invariants (rotation-proof), corroborated by explicit numerical rotations to $\sim 10^{-13}$.